# Prediction Models

# LogGOPSim - How It Works?

**Step 1: Tracing — `liballprof`**

- Intercepts MPI calls using **PMPI**
- Records:
    - MPI operations
    - start/end timestamps

**Step 2: GOAL Generation — `Schedgen`**

- Converts traces into a communication schedule
- Infers computation between MPI calls from timestamp differences

**Step 3: Parameter Measurement — `Netgauge`**

- Measures **LogGOP** parameters
- Used to get network model inputs for simulation

**Step 4: Simulation — `LogGOPSim`**

- Reads the generated schedule (bin file)
- Simulates the application using the measured values.

**Important note:**

- **Compile `liballprof` with MPICH**
- Then link/preload it with the MPI application to collect traces.

# References


1. https://spcl.inf.ethz.ch/Research/Performance/LogGOPSim/
2. Hoefler, Torsten, Timo Schneider, and Andrew Lumsdaine. "LogGOPSim: simulating large-scale applications in the LogGOPS model." *Proceedings of the 19th ACM International Symposium on High Performance Distributed Computing*. 2010.
3. Shen, Siyuan, et al. "Atlahs: An application-centric network simulator toolchain for ai, hpc, and distributed storage." *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. 2025.
4. Shen, Siyuan, et al. "Llamp: Assessing network latency tolerance of hpc applications with linear programming." *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*. IEEE, 2024.

# THANK YOU